

Dear examiner,

please find the criteria marks as well as teacher's comments below.

A: 4	B: 4	C: 3	D: 3	E: 5	Total: 19
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Criterion A: Presentation

A very comprehensive and logically developed IA. The student successfully introduces the mathematics required to investigate a real-life problem of his choice. Graphs, equations, tables and helpful diagrams are used effectively to aid understanding. The IA has an introduction with a clear set of aims, which are fully met. A conclusion summarizes the findings. There are no irrelevant or repetitive calculations.

Criterion B: Mathematical communication

Appropriate and consistent mathematical language is used at all times. The mathematical notation and communication is error-free. Unfamiliar key terms are defined and explained. A number of equations, diagrams, and tables are used.

Criterion C: Personal engagement

There is evidence of outstanding personal engagement. The student managed to really make this IA very unique and his own by investigating and applying the concepts required to solve his opening problem. All diagrams are made by the candidate himself; the Math software GeoGebra is used effectively.

Criterion D: Reflection

The student has done a very good job reflecting on his mathematical working at pretty much any stage which really helps to move the IA forward. Implications of the results are frequently discussed. The conclusion contains further critical reflection, a summary of his findings, and a discussion of limitations to his modeling.

Criterion E: Use of mathematics

All mathematical arguments are clear and logical. The student brings together and utilizes a variety of relevant mathematical concepts such as percentages, basic algebra, trigonometry, differential equations, Maclaurin series, limits, and complex numbers. All mathematical working is error-free. At times formulations like "relatively accurate", "better", "small" or "large" are not supported by quantitative explanations. The degree of accuracy used remains mostly unjustified.

Mathematics: Analysis and Approaches

Higher Level

Internal Assessment

Candidate Code: hdv625

**An investigation into the mathematics of a
pendulum**

Number of pages: 20

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1. Introduction, Rationale, and Aims

On TV I once saw a show that explained how a pendulum (see fig. 1) could be used to measure the velocity of a bullet. It had never occurred to me how the velocities of high-speed objects like bullets are actually measured. After learning about this application of a pendulum I wanted to develop an understanding of how and why a pendulum could be used for such applications, a topic that is heavily related to mathematics.

A pendulum consists of a weight, also known as a pendulum bob, that is suspended from a string. When a pendulum is displaced by a given angle, known as its angular displacement (θ), it moves back and forth around an equilibrium point. In physics class we learned that the pendulum's motion is an example of simple harmonic motion, meaning its motion is sinusoidal and therefore regardless of how far the pendulum is displaced its time period, the time the pendulum takes to complete one full cycle of motion (also known as an oscillation), will be constant. However, this conflicted with my personal experiences with pendulum-like systems, more specifically my use of swings as a child. I remember trying to get as high as possible to experience a sensation of floating, which quite clearly prolonged the time period of each of my swings. Upon further investigation I found that the idea that a pendulum's motion can be modelled by a sine curve is actually only based on an approximation that is effective for small angular displacements of the pendulum and in reality, a pendulum's time period does increase as its angular displacement increases (The Simple). I found this use of an approximation interesting and I wanted to understand why the approximation can be made, if it can be improved, and how it is useful. These are the focuses of this investigation.

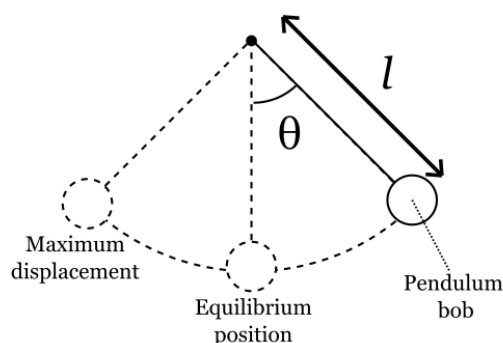


Figure 1. Basic diagram of a pendulum (Candidate's own image)

I have three main aims. Firstly, I will explain why a pendulum's motion can be approximately described as a sine curve for small angles and from this I will find an approximate expression for the time period of a pendulum, whose accuracy will be tested against an online database of exact pendulum time periods for various angular displacements. Secondly, I will derive an approximation for the time period of a pendulum that manages to more accurately estimate the pendulum's time period for larger angular displacements, an expression whose accuracy

will also be tested using the same online database. Thirdly, I will use the better approximation derived in this investigation to find an expression that approximates a sniper bullet's velocity based entirely on a pendulum's time period. The method used will be one particular approach of using a pendulum to approximate bullet velocity and will be discussed in detail later.

2. Differential equation describing the motion of a pendulum

Whilst this investigation will focus on the mathematics behind the pendulum, it is imperative to understand the physical principles that compel the pendulum to move as these govern the mathematics that describe the motion. To keep this brief, all that is critical to understand is that the pendulum moves because of an acceleration caused by gravity, denoted by $-g$ (Physics). Note that g is taken as negative because the acceleration of the pendulum will always be in the opposite direction to its displacement, i.e when the displacement of the pendulum is taken as positive, the acceleration is negative and vice versa. However, since the pendulum is attached to a string the component of gravity that acts on the pendulum is always tangential to its path (see fig. 2) and is therefore not $-g$, but rather

$$a = -g \sin \theta \quad (1)$$

where θ is the angular displacement of the pendulum from the vertical, and a is the acceleration of the pendulum. Looking at figure 2 this can be demonstrated using trigonometry.

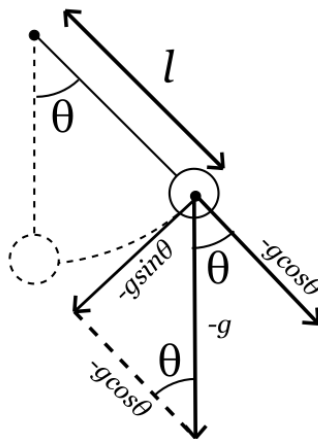


Figure 2. Diagram demonstrating the various accelerations acting on a pendulum (Candidate's own image)

Equation (1) is an expression for the acceleration of a pendulum. Nevertheless, realizing that the acceleration of the pendulum at any given θ can be given by equation (1) does not produce an equation that fully describes the motion. To do this one must find another expression for the pendulum's acceleration and equate it to equation (1). This can be done by finding the displacement of a pendulum along its arc at any given time and then differentiating twice

with respect to time to find the pendulum's acceleration. Firstly, to find the displacement of a pendulum along its arc it is helpful to imagine that the path of a pendulum can be extended to form an entire circle, which will have radius l (see fig. 3), the length from the pendulum bob's center of gravity to the pivot point of the pendulum. Then the area created by the vertical, the pendulum at a given displacement point, and the pendulum's path can be treated as a sector of this circle.

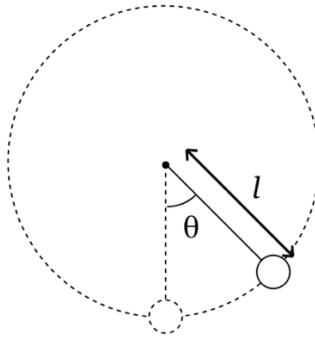


Figure 3. An extension of a pendulum's path to demonstrate that the displacement of a pendulum can be treated as a sector of a circle

(Candidate's own image)

Using figure 3 it becomes clear that the arc length formula can be applied to find the pendulum's displacement along its arc, giving the total displacement of the pendulum at a given angular displacement as:

$$x = l\theta \quad (2)$$

where l is the length of the pendulum, x its displacement from the equilibrium point along its arc, and θ is its angular displacement. Now that a displacement equation has been found it can be differentiated twice with respect to time to produce the desired equation for acceleration:

$$\frac{dx}{dt} = l \frac{d\theta}{dt} \quad (3)$$

$$a = \frac{d^2x}{dt^2} = l \frac{d^2\theta}{dt^2} \quad (4)$$

The expression in terms of the pendulum's length and angular acceleration can now be equated to the equation (1) to produce the following second order differential equation (since it contains a second derivative) describing the motion of a pendulum:

$$l \frac{d^2\theta}{dt^2} = -g \sin\theta \quad (5)$$

This can be simplified by rearranging and dividing through by l . This produces the final differential equation of motion for a pendulum which is as follows:

$$\frac{d^2\theta}{dt^2} + \frac{g}{l} \sin\theta = 0 \quad (6)$$

3.1. The limitations of the differential equation for a pendulum's motion

Whilst equation (6) does accurately describe the motion of a pendulum, it is very difficult to solve by standard methods. However, to find an expression for the time period of a pendulum, one of my ultimate goals, a solution to equation (6) must be found, since an expression for θ in terms of time would allow one to find the time it takes for a pendulum to complete one oscillation, the time period. If one tries to solve equation (6) analytically for θ , a non-solvable integral known as an elliptic integral is produced, something that goes well beyond this investigation. The reason equation (6) is so challenging to solve is because it is a non-linear differential equation, meaning the difference between the outputs produced by two different solutions is not proportional to the difference in the inputs to garner those two solutions (Abdelkader). This leads to an equation that can become unpredictable and chaotic. Equation (6) can be clearly shown to be non-linear. Substituting three different, evenly spread values for θ , say $\frac{\pi}{4}$, $\frac{\pi}{2}$, and $\frac{3\pi}{4}$ will provide solutions $\frac{d^2\theta}{dt^2} = -\frac{\sqrt{2}g}{2l}$, $-\frac{g}{l}$, and $-\frac{\sqrt{2}g}{2l}$ respectively. In the case of a linear differential equation, a constant difference between two inputs would always produce two solutions with a constant difference when subtracted from one another. This can be tested with our arbitrarily selected values. The difference between the first two inputs for θ is $\frac{\pi}{4} - \frac{\pi}{2} = -\frac{\pi}{4}$, and the difference between the second two inputs for θ is $\frac{\pi}{2} - \frac{3\pi}{4} = -\frac{\pi}{4}$. The difference between the solutions from the first two inputs ($\frac{\pi}{4}$ and $\frac{\pi}{2}$) is $-\frac{\sqrt{2}g}{2l} + \frac{g}{l} = \frac{g}{l}(\frac{2-\sqrt{2}}{2})$ and the difference between the solutions produced by the second two inputs ($\frac{\pi}{2}$ and $\frac{3\pi}{4}$) is $-\frac{g}{l} + \frac{\sqrt{2}g}{2l} = \frac{g}{l}(\frac{-2+\sqrt{2}}{2})$. Quite clearly a difference of $-\frac{\pi}{4}$ between any θ inputs does not produce a constant difference between the respective solutions from these inputs, proving that equation (6) is a non-linear differential equation. This non-linear equation arises because of the presence of $\sin\theta$. If $\sin\theta$ could be approximated by a function that is linear, then for a given constant difference between two inputs, the difference in outputs produced by these inputs would also be constant. As previously mentioned, non-linear differential equations are extremely challenging to solve, whereas linear ones are typically far more easily analytically solved. Whilst it is not yet clear why a differential equation being linear is beneficial, it will become clear once the equation (6) is successfully linearized, the next step in this investigation.

3.2. Linearizing the differential equation by approximating $\sin\theta = \theta$

Finding an expression that linearizes the function $\sin\theta$ can be done by expressing said function in the form of a polynomial, and then approximating $\sin\theta$ to the part of the polynomial that behaves like a linear function. To do this one must first consider the Maclaurin series expansion which states that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \quad (7)$$

Applying equation (7) to $\sin\theta$ by setting $f(x) = \sin\theta$ produces the following first 4 terms going up to $n = 3$:

$$\sin\theta = \frac{\sin(0)}{0!}\theta^0 + \frac{\cos(0)}{1!}\theta^1 + \frac{-\sin(0)}{2!}\theta^2 + \frac{-\cos(0)}{3!}\theta^3 + \dots \quad (8)$$

$$\sin\theta = \theta - \frac{1}{6}\theta^3 + \dots \quad (9)$$

Using this polynomial describing $\sin\theta$ from equation (9), a linear equation approximating $\sin\theta$ can be found by simply taking the first term and disregarding all terms afterwards to therefore approximate $\sin\theta = \theta$. Although taking the polynomial to only one term decreases accuracy, this must be done since taking any more terms and substituting them into equation (6) would once again produce a non-linear differential equation. The validity of this approximation can be further demonstrated by graphing both $f_1(\theta) = \sin\theta$ and $f_2(\theta) = \theta$ (see fig. 3).

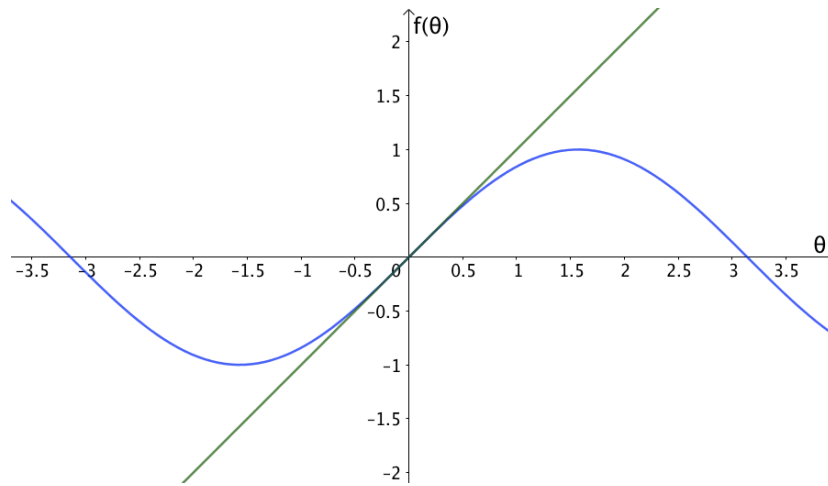


Figure 4. Curves $f_1(\theta) = \theta$ (green) and $f_2(\theta) = \sin\theta$ (blue) (Candidate's own image)

What becomes clear from figure 4 is that as $\sin\theta$ approaches 0 so does θ , demonstrating that for values of θ close to zero this approximation is fairly accurate. The extremely similar values for θ that both functions produce near 0 can further be demonstrated by claiming that

$$\lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} = 1 \quad (10)$$

This limit suggests that as θ approaches 0 the values for $\sin\theta$ and θ become closer and closer to each other and therefore when divided by one another approach 1. The validity of this limit cannot be immediately be confirmed since substituting $\theta = 0$ would produce the fraction $\frac{0}{0}$ which is undefined. Nevertheless, the validity can be confirmed by applying l'Hospital's rule to differentiate both the numerator and denominator of the fraction with respect to θ to produce the following:

$$\lim_{\theta \rightarrow 0} \frac{\cos(\theta)}{1} \quad (11)$$

Now that a fraction with a denominator of 0 will not be produced, 0 can be substituted proving that this limit is in fact equal to 1:

$$\frac{\cos(0)}{1} = 1 \quad (12)$$

From this it has been proved that for values of θ closer to 0 the approximation becomes increasingly accurate and at 0 the approximation $\sin\theta = \theta$ is exact.

3.3. Analytically solving the differential equation for the motion of a pendulum using $\sin\theta = \theta$

Now that a linear approximation for $\sin\theta$ has been found it can be substituted into equation (6) to produce a linear differential equation:

$$\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0 \quad (13)$$

Whilst it may not be clear why this simplification is helpful at first, once one dissects what the equation is saying it becomes clear why linear differential equations are so desirable. To better understand the equation firstly rearranging as follows is helpful:

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\theta \quad (14)$$

Equation (14) simply states that θ multiplied by the constant $-\frac{g}{l}$ equals its second derivative with respect to time.

Understanding this one realizes that such a function that possesses this recursive feature is e^t since any derivative of e^t with respect to time simply equals e^t . However, setting θ equal to e^t does not satisfy the equation due to the constant $-\frac{g}{l}$. Instead the t in the exponent must be multiplied by a constant that accounts for this constant so that when the expression is differentiated with respect to time twice it equals to its initial input. Additionally, the expression that can be set for θ using e^t can be multiplied by any constant since this will have no impact on the relationship between θ and its second derivative. Understanding this one can set

$$\theta = C_1 e^{rt} \quad (15)$$

where C_1 and r are constants, producing the following:

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}C_1 e^{rx} \quad (16)$$

Since we have assumed $\theta = C_1 e^{rt}$, a value for the second derivative for θ can be found by differentiating twice with respect to time:

$$\frac{d\theta}{dt} = C_1 r e^{rt} \quad (17)$$

$$\frac{d^2\theta}{dt^2} = C_1 r^2 e^{rt} \quad (18)$$

The expression for $\frac{d^2\theta}{dt^2}$ can also be substituted into the original differential equation

$$C_1 r^2 e^{rt} = -\frac{g}{l} C_1 e^{rt} \quad (19)$$

This process could theoretically be done with any linear differential equation which is why they are so desirable (Second-Order). Because the change in inputs is proportional to the change in outputs, a function and all of its derivatives will have a similar recursive relationship to the function e^t . As seen in equation (19), this produces a result where both sides can be divided by $C_1 e^{rt}$ and therefore a solution for r can be found. Using the solution for r an analytical solution for equation (13) can be found, since once r is found it can be substituted into equation (15) to find a function for θ that solves equation (13). To do this one must firstly divide equation (19) by $C_1 e^{rt}$ and rearrange for r as follows:

$$r^2 = -\frac{g}{l} \quad (20)$$

$$r = \pm \sqrt{-\frac{g}{l}} \quad (21)$$

$$r = \pm i \sqrt{\frac{g}{l}} \quad (22)$$

As equation (22) involves a radical and a fraction, it can look somewhat messy when put into larger expressions.

To avoid confusion, it will be assumed moving forward that

$$\sqrt{\frac{g}{l}} = w \quad (23)$$

Now that two values for r have been found in equation (22) they can be substituted into the initial assumption for θ that was used to solve this differential equation. Doing so produces two different solutions to the differential equation

$$\theta = C_1 e^{iwt} \quad (24)$$

$$\theta = C_2 e^{-iwt} \quad (25)$$

Initially it appears unusual that two general solutions have been found, given that the general solution should encompass all possible solutions. However, these two solutions are in fact not the most general solution to the equation. Instead the sum of the two solutions is. To understand why this is one must understand that a second order differential equation is an equation describing the rate of change of a derivative. In the context of a pendulum

the differential equation states the rate of change of its velocity. Given the acceleration of a pendulum it is only possible to know the position of the pendulum at a given point in time if you are given two details about the pendulum: its velocity at a specific point in time and its position at a specific point in time. To include both of these pieces of information one must produce a general solution that has two constants. Since our differential equation is linear these constants can be added together (Second-Order), giving the following:

$$\theta = C_1 e^{i\omega t} + C_2 e^{-i\omega t} \quad (26)$$

Before applying boundary conditions that will help find a particular solution to the equation of motion of a pendulum, equation (26) can be simplified by converting from Euler's form into Polar form

$$\theta = C_1(\cos(\omega t) + i\sin(\omega t)) + C_2(\cos(\omega t) - i\sin(\omega t)) \quad (27)$$

$$\theta = C_1 \cos(\omega t) + C_1 i\sin(\omega t) + C_2 \cos(\omega t) - C_2 i\sin(\omega t) \quad (28)$$

$$\theta = (C_1 + C_2) \cos(\omega t) + (iC_1 - iC_2) \sin(\omega t) \quad (29)$$

Since C_1 and C_2 are both constants this can be written more simply as

$$\theta = A \cos(\omega t) + B \sin(\omega t) \quad (30)$$

where A and B are constants. Equation (30) acts as the overall general solution to the differential equation. From here boundary conditions can be applied. Whilst any boundary conditions could be selected, the most intuitive boundary conditions are firstly that when $t = 0$, $\theta = \theta_o$ (where θ_o is the maximum angular displacement of the pendulum). Additionally, when imagining the motion of the pendulum it can immediately be realized that at θ_o the pendulum is changing directions and as a result $\frac{d\theta}{dt} = 0$. Therefore, our two boundary conditions are that when $t = 0$, $\theta = \theta_o$ and $\frac{d\theta}{dt} = 0$. Incorporating the first boundary condition produces the following:

$$\theta_o = A \cos(0) + B \sin(0) \quad (31)$$

$$\theta_o = A \quad (32)$$

Now, in order to incorporate the second boundary condition, the general solution from equation (30) can be differentiated with respect to time which produces:

$$\frac{d\theta}{dt} = -A \sin(\omega t) + B \cos(\omega t) \quad (33)$$

Substituting in the second boundary conditions produces

$$0 = -A \sin(0) + B \cos(0) \quad (34)$$

$$0 = B \quad (35)$$

Now that values for A and B have been found a particular solution to the differential equation that describes the motion of the pendulum is as follows:

$$\theta = \theta_o \cos(wt) \quad (36)$$

3.4. Finding the time period of the pendulum

Whilst our differential equation has been solved to satisfy the motion of the pendulum, its time period is still yet to be found. Nevertheless, this can easily be found by noticing that the particular solution, equation (36), is a cosine curve. Arbitrarily selecting values for θ_o and w this can be shown. Setting $\theta_o = \frac{\pi}{6}$ and $w = 1$, the following curve is created (see fig. 5).

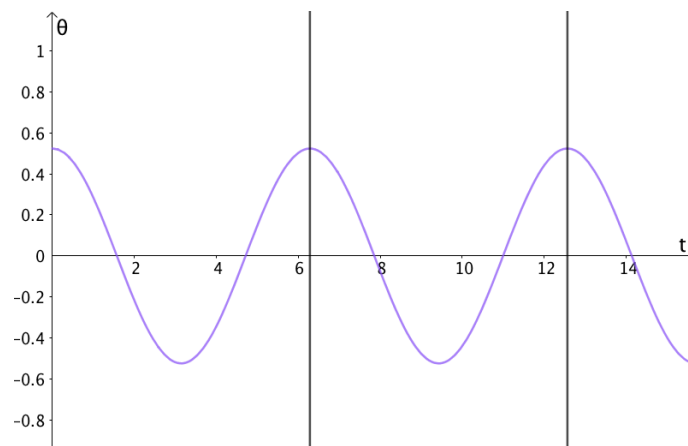


Figure 5. Graph depicting the curve $\theta = \frac{\pi}{6}\cos(t)$ (purple) where vertical lines are at $t = 2\pi$ and $t = 4\pi$ respectively (Candidate's own image)

From figure 5 it can be deduced that the time period for this particular system is 2π , since this is the time it takes for the sine curve and the pendulum to complete one full oscillation. However, a more general time period can be deduced once it is understood that the period of a sine/cosine curve is equal to $\frac{2\pi}{b}$, where b is the coefficient of input into the sine curve, in our case t (Amplitude). Since w acts as the constant coefficient of t in the particular solution, a formula for the time period can finally be deduced

$$T = \frac{2\pi}{w} \quad (37)$$

Rewriting w in terms of the individual constants it contains finally produces the formula

$$T = 2\pi \sqrt{\frac{l}{g}} \quad (38)$$

This is the final time period for a pendulum, assuming $\sin(\theta) = \theta$.

3.5. Evaluating accuracy of time period expression

Whilst equation (38) makes it appear as if angular displacement of a pendulum plays no role a pendulum's time period, one must remember that this is only an approximate expression, and the value of θ will impact the time period of a real pendulum. Using an online database (Large), the percentage error of the expression for time period can be calculated for any given angle of θ . To calculate percentage error the following equation will be used

$$\text{Percentage error in } T = \frac{|T_{\text{real}} - T_{\text{approximate}}|}{T_{\text{real}}} \times 100 \quad (39)$$

where T_{real} is the actual time period of a pendulum for a given angular displacement and $T_{\text{approximate}}$ is the approximate time period given by equation (38). Once again, for simplicity, both l and g will be taken as 1. From here the accuracy of the time period formula found will be tested for 5 arbitrarily selected values of θ .

Maximum angular displacement (θ)	$T_{\text{approximate}}$	T_{real} (to 6 significant figures)	Percentage error (to three significant figures)
$\frac{\pi}{16}$	2π	6.29835	0.241
$\frac{\pi}{8}$	2π	6.34427	0.972
$\frac{\pi}{4}$	2π	6.53398	3.99
$\frac{\pi}{2}$	2π	7.38910	17.6
$\frac{5\pi}{6}$	2π	10.0315	59.7

As can be seen from the data, for smaller angles the time period approximation is very accurate, but as angular displacement increases, so does the error in the approximation. This is to be expected given the limit in equation (10). The fact that the approximate time period stays constant dictates the fact that it becomes increasingly less accurate the larger the angular displacement, since larger angular displacements increase a pendulum's time period.

4.1. Finding an approximation for $\sin(\theta)$ that increases accuracy in time period expression for larger angular displacements

The applications of pendulums far exceed pendulums that are released from small angles, and as seen from the error analysis of the time period in equation (38), the approximation found becomes increasingly less representative of a pendulum's motion the larger the angular displacement becomes. It is entirely situational when

the approximation for $\sin\theta = \theta$ is valid enough to use for a pendulum system and for a given engineering project involving a pendulum. The approximation's helpfulness depends on various factors like how accurate the calculations must be, how far the pendulum is being displaced, and the goal one is trying to achieve with the pendulum. Nevertheless, it is certainly true to $\sin\theta = \theta$ will not suffice for all real-life situations. This begs the question of whether there is a way to find an approximation that better expresses the motion of a pendulum, and is in particular more accurate for larger angular displacements. To do this it seems sensible to start off analysing the initial approximation that we made and the reason for its limitations.

The approximation of $\sin(\theta) = \theta$ overestimates the value of $\sin(\theta)$ for larger angles that get further away from zero. This becomes visually obvious when comparing how representative the linear approximation θ is of the curve $\sin(\theta)$ the larger the angle becomes:

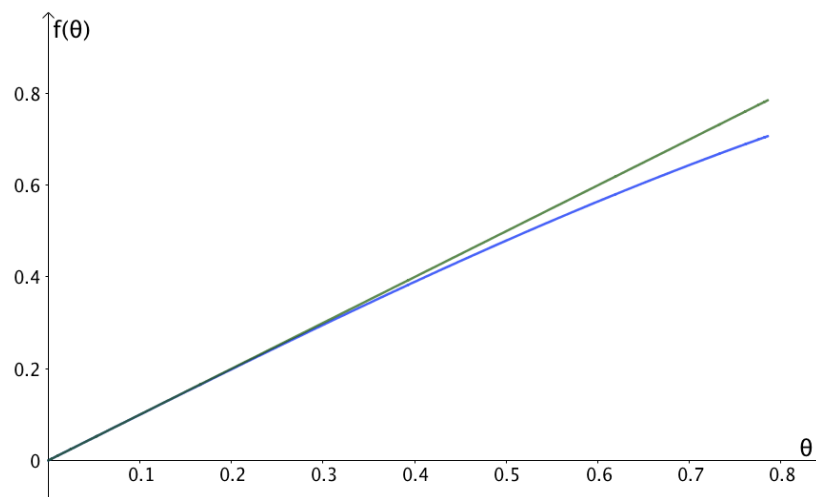


Figure 6. Curves $f_1(\theta) = \theta$ (green) and $f_2(\theta) = \sin\theta$ (blue) for $0 \leq \theta \leq \frac{\pi}{4}$ (Candidate's own image)

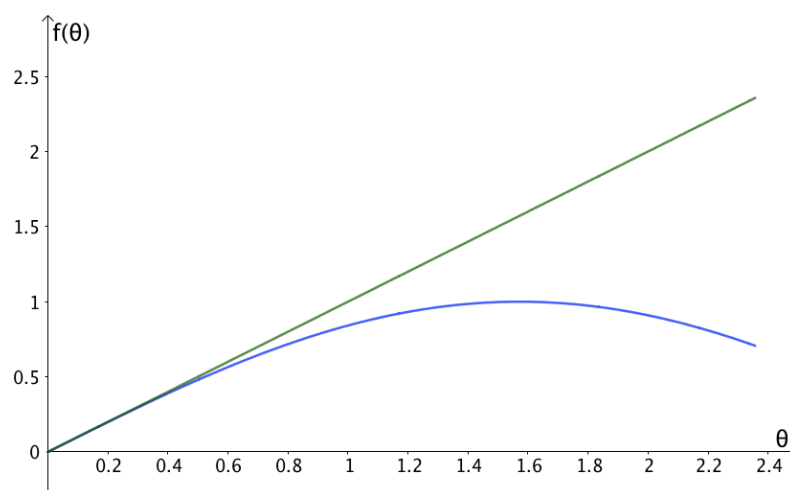


Figure 7. Curves $f_1(\theta) = \theta$ (green) and $f_2(\theta) = \sin\theta$ (blue) for $0 \leq \theta \leq \frac{3\pi}{4}$ (Candidate's own image)

What becomes clear from figure 6 and figure 7 is that to increase accuracy in the approximation for larger angles the gradient of $f_1(\theta)$ must become shallower, whereas for smaller angles keeping the gradient close to identical

would be ideal given the already very accurate linear approximation. Because of this we seek a correction term that we can multiply the approximation $\sin(\theta) = \theta$ by such that when θ is near 0 the term nears 1 and that as θ approaches π the correction term approaches 0. Interestingly such a term can be found when one looks at the double angle trigonometric identity for $\sin\theta$ (Millet):

$$\sin 2\theta = 2\sin\theta\cos\theta \quad (40)$$

Dividing the angle by two to give an equation for $\sin\theta$ gives the following:

$$\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} \quad (41)$$

First and foremost, $\sin\frac{\theta}{2}$ can be approximated by $\frac{\theta}{2}$, since $\frac{\theta}{2}$ can simply be substituted for θ in the Maclaurin series in equation (9). This acts as the first step to find a linear approximation for $\sin\theta$. Doing this produces

$$\sin\theta = \theta\cos\frac{\theta}{2} \quad (42)$$

From here it appears we have reached a dead-end, since, although this expression is more accurate, it is not linearized. However, it can be linearized by making $\cos\frac{\theta}{2}$ a constant. Firstly, the property that as θ approaches π , $\cos\frac{\theta}{2}$ approaches 0 must remain so the correction term behaves as intended. Because of this $\cos\frac{\theta}{2}$ must still depend on an angle. Cleverly, to make this a constant one can consider that for any given pendulum system its maximum angular displacement is a constant. This means that we can approximate $\cos\frac{\theta}{2}$ as follows:

$$\cos\frac{\theta}{2} = \cos\frac{\theta_o}{2} \quad (43)$$

where θ_o is the pendulum's maximum angular displacement. This approximation works as intended. As θ_o approaches π the term approaches 0 and for small values of θ_o the term is close to 1. Substituting equation (43) into the $\sin\theta$ approximation produces the final approximation

$$\sin\theta = \theta\cos\frac{\theta_o}{2} \quad (44)$$

This approximation can be put to the test by graphing the linear approximations for various values of θ_o . From figures 8 and 9 (below) it becomes clear that this approximation allows for a more accurate linear approximation of $\sin\theta$ than the old approximation $\sin\theta = \theta$.

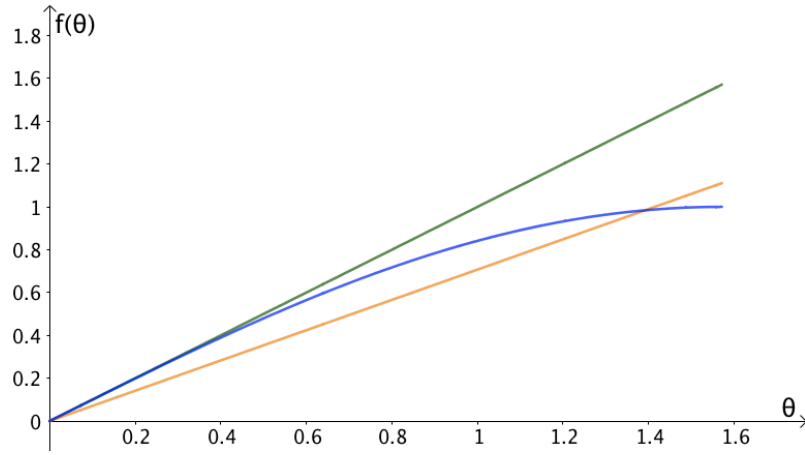


Figure 8. Curves $f_1(\theta) = \theta$ (green), $f_2(\theta) = \sin\theta$ (blue), and $f_3(\theta) = \theta\cos\frac{\pi}{4}$ (orange) for $0 \leq \theta \leq \frac{\pi}{2}$ (Candidate's own image)

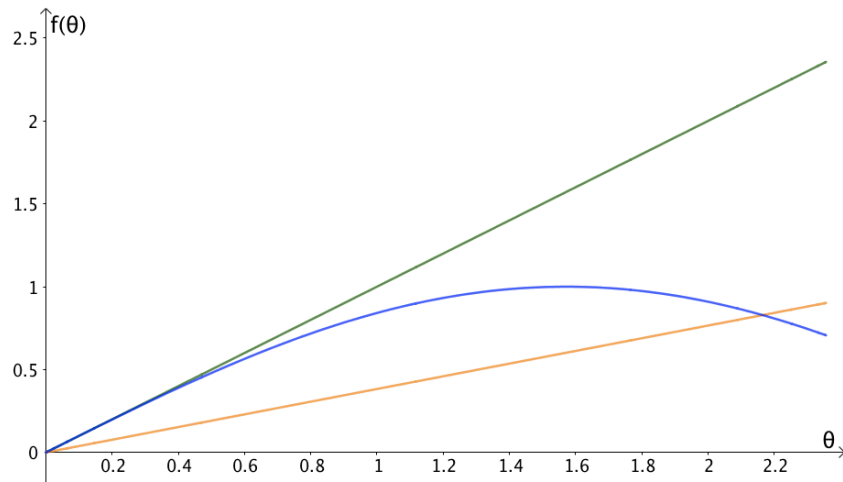


Figure 9. Curves $f_1(\theta) = \theta$ (green), $f_2(\theta) = \sin\theta$ (blue), and $f_3(\theta) = \theta\cos\frac{3\pi}{8}$ (orange) for $0 \leq \theta \leq \frac{3\pi}{4}$ (Candidate's own image)

Given the increased accuracy of this approximation for $\sin\theta$ it can subsequently replace the term $\sin\theta$ in equation (6) to linearize the equation of motion for a pendulum

$$\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta\cos\frac{\theta_0}{2} = 0 \quad (45)$$

The equation produced is once again linearized since $\cos\frac{\theta_0}{2}$ is a constant. Whilst all the steps to find the solution to equation (45) can be done, it is unnecessary since it would be the exact same process as solving the equation without the term $\cos\frac{\theta_0}{2}$. Therefore, based on the previous solution of the differential equation where $\sin\theta = \theta$, a similar solution can also be found treating $\cos\frac{\theta_0}{2}$ as another constant. This produces the following general solution to equation (45):

$$\theta = A \cos\left(\left(\sqrt{\cos\frac{\theta_0}{2}}\right)wt\right) + B \sin\left(\left(\sqrt{\cos\frac{\theta_0}{2}}\right)wt\right) \quad (46)$$

The particular solution based on the same boundary conditions as before is therefore:

$$\theta = \theta_o \cos \left(\left(\sqrt{\cos \frac{\theta_o}{2}} \right) \omega t \right) \quad (47)$$

Once again applying the rule that the period of the sine curve produced in equation (47) equals the time period of a pendulum the approximation for the time period can be improved and now written as

$$T = 2\pi \sqrt{\frac{l}{g}} \times \frac{1}{\sqrt{\cos \frac{\theta_o}{2}}} \quad (48)$$

which can then be simplified to

$$T = 2\pi \sqrt{\frac{l}{g \cos \frac{\theta_o}{2}}} \quad (49)$$

This is the final time period for a pendulum for the improved approximation $\sin \theta = \theta \cos \frac{\theta_o}{2}$. The domain in which it is valid for a pendulum is $0 < \theta < \pi$.

4.2. Evaluating accuracy of the improved time period expression

Similar to before, the accuracy of the time period found in equation (49) will be tested using the percentage error calculation from equation (39). It will be done under the same conditions, where both l and g are equated to 1. The time periods will be tested against the same online database (large).

Maximum angular displacement (θ)	$T_{approximate}$ (to 6 significant figures)	T_{real} (to 6 significant figures)	Percentage error (to three significant figures)
$\frac{\pi}{16}$	6.29837	6.29835	0.000318
$\frac{\pi}{8}$	6.34443	6.34427	0.00252
$\frac{\pi}{4}$	6.53691	6.53398	0.0448
$\frac{\pi}{2}$	7.47201	7.38910	1.12
$\frac{5\pi}{6}$	12.3504	10.0315	23.1

Similarly, to the first time period approximation, the smaller the maximum angular displacement is the more accurate the time period is. However, the overall accuracy for every single maximum angular displacement has

increased in comparison to the first time period, suggesting this approximation is better. This is because, unlike before, this approximation considers the angular displacement of the pendulum which, as seen in the data, effects the pendulum's time period. One thing that must be noted is that the percentage errors are likely not exact due to rounding in both the values. Also, one interesting observation is that for the improved approximation the time period is overestimated, whereas before it was underestimated. Regardless of this the new found approximation for a pendulum's time period is clearly more accurate for all angular displacements, reflected by the universally smaller percentage errors for each of the angular displacement values.

5. Real life application of a pendulum using its time period

As mentioned before the reason I initially wanted to investigate the pendulum was because of a TV show that showed how a pendulum could be used to measure bullet velocity. Whilst they showed the pendulum measuring bullet velocity, they did not explain the mathematics behind it all. Therefore, I will mathematically determine a way in which a bullet's velocity can be measured by a pendulum using the improved time period approximation found in equation (49), a process that will be explained soon. However, before delving into the mathematics, the basic premise of this real-life situation must be explained. The idea is that a bullet is shot at a pendulum and upon contact lodged into the pendulum. After the bullet makes contact, the pendulum is forced into motion and its time period is measured, which can then be used to find bullet velocity. The goal of this real life problem is twofold: firstly, I will determine an expression that gives an approximate value for the velocity of a bullet based entirely on a pendulum's time period. Secondly, I will apply said formula to a sniper bullet, the .308 Winchester, to evaluate the approximate expression found for bullet velocity.

The first step to finding an expression for the velocity of a bullet in terms of a pendulum's time period is to find an expression for the maximum velocity of a pendulum. Using the laws of momentum, an expression for bullet velocity can be found using the maximum velocity of a pendulum. As this is not a physics investigation all that must be known is that *Momentum = Mass × Velocity* and in a given system momentum must always be conserved. In the context of the pendulum this means that when the bullet is fired at the pendulum the entire momentum of the bullet will be transferred into the pendulum when the bullet lodges itself in the pendulum. This can be expressed mathematically as follows:

$$M_B \times V_B = (M_P + M_B) \times V_{max} \quad (50)$$

M_B and M_P are the masses of the bullet and pendulum respectively, V_B is bullet velocity, and V_{max} is the maximum velocity of a pendulum, i.e. its velocity at its equilibrium point, which is the point where the bullet would make

contact with the pendulum. Whilst equation (50) provides an expression for bullet velocity, it does so using a velocity measurement for the pendulum. In theory this would be far harder to measure than the pendulum's time period because the pendulum's velocity changes instantaneously. As a result, V_{max} must be rewritten in terms of the pendulum's time period, a process that can be achieved using equation (49) found earlier in this investigation. This can be achieved because equation (49) considers the angular displacement of a pendulum, hence if an expression for V_{max} is found in terms of angular displacement it can also be found in terms of the approximate pendulum time period. At this point it becomes clear why finding more accurate approximations is helpful. This method for finding bullet velocity would be impossible if one merely assumed $\sin\theta = \theta$ because this approximation fails to consider different angular displacements. Coming back to the situation, writing V_{max} in terms of angular displacement requires an equation that mathematically relates a pendulum's potential energy to its kinetic energy. Once again, since this is not a physics investigation, all one must understand is that energy in a pendulum system energy is always conserved and a pendulum's maximum kinetic energy is equal to its maximum potential energy, something that can be expressed as follows:

$$mgh_{max} = \frac{1}{2}m(V_{max})^2 \quad (51)$$

where m is the mass of the pendulum, and h_{max} is the maximum vertical displacement of a pendulum from its equilibrium point. Now equation (51) can be rearranged to find an expression for V_{max} by dividing both sides by m

$$gh_{max} = \frac{1}{2}(V_{max})^2 \quad (52)$$

$$2gh_{max} = (V_{max})^2 \quad (53)$$

$$V_{max} = \sqrt{2gh_{max}} \quad (54)$$

From here h_{max} can be written in terms of the pendulum's angular displacement, something that can be found using trigonometry (see fig. 10).

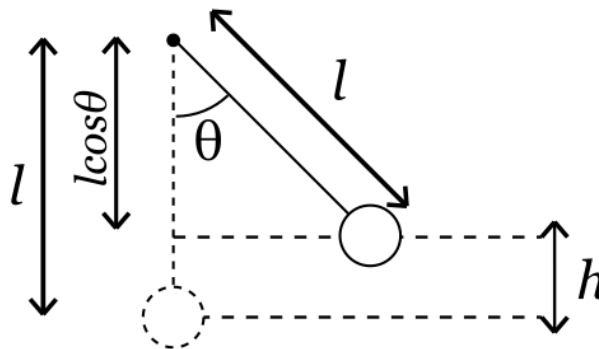


Figure. 10. Finding the max height of a pendulum using trigonometry (Candidate's own image)

From figure 10 it is clear the max height of the pendulum is as follows:

$$h_{max} = L - L\cos\theta_0 \quad (55)$$

$$h_{max} = L(1 - \cos\theta_0) \quad (56)$$

This expression can be resubstituted into maximum velocity formula as follows:

$$V_{max} = \sqrt{2gL(1 - \cos\theta_0)} \quad (57)$$

Having almost achieved the goal of finding V_{max} in terms of a pendulum's time period, one should be reminded of our improved approximation for time period in equation (49):

$$T = 2\pi \sqrt{\frac{l}{g\cos\frac{\theta_0}{2}}} \quad (58)$$

This formula can be rearranged for θ_0 , producing a formula for θ_0 in terms of T . This can subsequently be substituted into the V_{max} equation to find an approximation for V_{max} in terms of time period. It must be noted that at this point the exact expression for V_{max} is being transformed into an approximation, since the time period expression that is being rearrange and substituted is an approximation. To find an expression for θ_0 in terms of T both sides of equation (58) can be squared:

$$T^2 = 4\pi^2 \frac{l}{g\cos\frac{\theta_0}{2}} \quad (59)$$

Then one can rearrange for $\cos\frac{\theta_0}{2}$

$$\frac{T^2 g}{4\pi^2 l} = \frac{1}{\cos\frac{\theta_0}{2}} \quad (60)$$

$$\cos\frac{\theta_0}{2} = \frac{4\pi^2 l}{T^2 g} \quad (61)$$

Finally θ_0 can be deduced as follows

$$\frac{\theta_0}{2} = \arccos\left(\frac{4\pi^2 l}{T^2 g}\right) \quad (62)$$

$$\theta_0 = 2\arccos\left(\frac{4\pi^2 l}{T^2 g}\right) \quad (63)$$

Now equation (63) can be plugged into the V_{max} from equation (57) formula as previously mentioned:

$$V_{max} = \sqrt{2gL(1 - \cos\left(2\arccos\left(\frac{4\pi^2 l}{T^2 g}\right)\right))} \quad (64)$$

The final step to find an expression for bullet velocity in terms of the pendulum's time period is to simply substitute equation (64) into the conservation of momentum formula for our pendulum system. Doing so gives:

$$M_B \times V_B = (M_P + M_B) \times \sqrt{2gL(1 - \cos \left(2\arccos \left(\frac{4\pi^2 l}{T^2 g} \right) \right))} \quad (65)$$

$$V_B = \frac{(M_P + M_B) \times \sqrt{2gL(1 - \cos \left(2\arccos \left(\frac{4\pi^2 l}{T^2 g} \right) \right))}}{M_B} \quad (66)$$

With equation (66) it now becomes possible to estimate the velocity of a bullet by only knowing the time period of the pendulum (assuming one knows the mass of the bullet prior to the experiment). Nevertheless, one must give all the constants appropriate values which we will now do under the assumption that a .308 Winchester sniper bullet is being used. Firstly, M_B will be taken as 0.00972kg, the approximate weight of a .308 Winchester bullet (.308), and g will be taken as 9.81ms^{-2} since this is the acceleration of gravity on Earth's surface. In the case of both M_P and L , in ideal circumstances these would be as large as possible. This is because the larger these values are, the less the pendulum would be displaced and, as shown in section 4.2, reducing the maximum angular displacement of a pendulum increases the accuracy of our time period approximation from equation (49) which is being applied here. Nevertheless, it is unrealistic to take M_P and L as extremely high values, hence they will arbitrarily be assumed to be 10kg and 2.5m respectively to help further this investigation. To ensure these values are realistic, all constants will be substituted and tested to see if a time period can be found for just above the largest potential velocity of any sniper bullet, 1500ms^{-1} (.220), because if it cannot be then that means the bullet at maximum velocity would supply enough energy to the pendulum to make it swing over 180° , making our time period approximation not applicable. Plugging in all these constants into equation (66) gives:

$$1500 = \frac{(10.00972) \times \sqrt{49.05(1 - \cos \left(2\arccos \left(\frac{10\pi^2}{9.81T^2} \right) \right))}}{0.00972} \quad (67)$$

From this a solution for the time period can be found which is as follows:

$$T \approx 3.189 \quad (68)$$

Although it has been confirmed that the pendulum will not swing more than 180° when bullet velocity is highest, it is still useful to verify that time period found in equation (68) is a period for a relatively low maximum angular displacement since lower angular displacements increase accuracy in calculations. This can be verified by substituting the time period found in equation (68) and the other relevant constants into equation (49) to find the maximum angular displacement for our pendulum given its theoretical conditions. The maximum angular displacement found is as follows:

$$3.189 = 2\pi \sqrt{\frac{2.5}{9.81 \cdot \cos \frac{\theta_0}{2}}} \quad (69)$$

$$\theta_0 \approx 16.8^\circ \quad (70)$$

Given this relatively low maximum angular displacement for our system given its theoretical constants, one can be confident that equation (67) will provide relatively accurate bullet velocity approximations and therefore equation (67) can be graphed, producing the following.

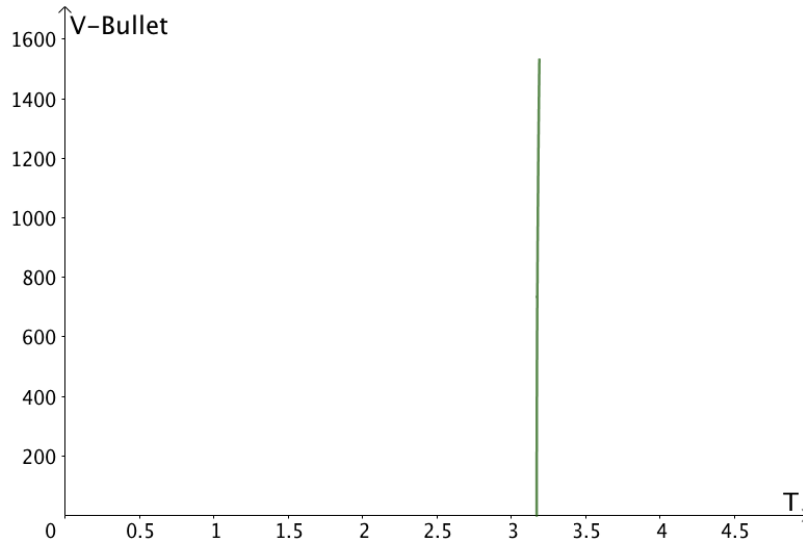


Figure 11. Curve $V_B = \frac{(10.00972) \times \sqrt{49.05(1 - \cos(2 \arccos(\frac{10\pi^2}{9.8172}))}}{0.00972}$ (green) for $0 \leq T \leq 3.189$. Note that the y-axis label is not V_B due to

GeoGebra software limitations. (Candidate's own image)

Figure 11 shows that the function V_B for the given domain is almost vertical. This should in fact have been expected since the variations in time period of a pendulum are very small between different angular displacements in comparison to the variations in bullet velocity. Therefore, for any constants selected that are realistic for the given real-life situation, the graph produced for V_B will be almost vertical. This does pose the problem that to find a strong approximation for bullet velocity, it is required to have an incredibly precise reading for the pendulum's time period, since an imprecise reading will likely hugely change the approximate bullet velocity found. When considering this in conjunction with the fact that air and frictional resistances were not considered, it points to the following real-life approximation found only being theoretically useful but actually to be somewhat useless in real life. A potential improvement would be to directly measure the angular displacement using high resolution photography, instead of measuring for time period, since the larger variations in angular displacement would require less precise (albeit still very precise) readings, which in the real world is more realistic.

Conclusion and Reflection:

This exploration dissected the motion of a pendulum and explored how the second order differential equation of motion for a pendulum can be linearized and subsequently solved. From the differential equation's solution an approximation for the time period of a pendulum was found and then tested against exact values for the time period of a pendulum for various maximum angular displacements. From this it became clear that the approximation was only valid for small angles, reducing its real-world applications. Therefore a better approximation to linearize the differential equation was found and used to find a more accurate approximation for the time period of a pendulum for larger angles, which was proven to be more accurate than the first through database comparisons. Finally, the result of the more accurate time period approximation was used to explore how a pendulum could measure the velocity of a bullet, which was then tested using the .308 Winchester sniper bullet.

When analysing the results of this investigation, it is important to consider the investigation's limitations. Firstly, neither of the approximations found for the time period effectively approximated the time period of angles above $\frac{\pi}{2}$ (both above 1% error), although the second approximation is less inaccurate. To add to the investigation, exploring other ways to find more accurate approximations than $\sin\theta = \theta$ would lead to a better understanding of different thought processes used to derive approximations. Also, this investigation was reliant on pre-existing knowledge of the actual time periods of a pendulum to measure the accuracy of the approximations uncovered, a luxury not always experienced in real life. To further this investigation, it would be interesting to try and derive the exact formula describing the time period of a pendulum.

In terms of the real-world application of using a pendulum to measure bullet velocity there were also various limitations. Firstly, the entire real-life situation relied upon very specific conditions and assumptions, such as no air resistance, that simply would not hold in real life. This forces one to question how valid using our approximation for time period would be if these assumptions and conditions were not true. In addition, the final expression found for bullet velocity purely in terms of time period requires one to input an incredibly accurate time period to receive a reasonable approximation for bullet velocity. This acts as a paradox to the initial purpose of finding the approximation for time period, since when one attempts to find an approximation one is truly attempting to find a simplification. However, for the approximation of bullet velocity to be valid one requires a value for time period that is so precise and accurate that the situation in many ways becomes more complicated, forcing one to question whether the approximation actually led to a simplification. Nevertheless, the complete analysis I performed did enable me to propose an alternative method of measuring the maximum displacement of the pendulum to find the bullet velocity, which would require less precise readings to be effective.

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